Title: Macroscopic Coherence: The Thermodynamic Evolution of Human Awareness Across Historical and Anthropological Epochs
Author: Nickolas Patrick Joseph Schoff **Affiliation:** Schoff Research Program **Date:** June 2, 2026

Abstract

Historical and anthropological models traditionally view the evolution of human consciousness as a linear accumulation of cognitive capacity. This paper proposes a structural revision, modeling historical epochs as thermodynamic phase transitions governed by the Computational Surplus threshold ($C_s = R_{obs} - R_{env}$). By integrating the Constraint-Relaxation Energy Model (CREM) and Bidirectional Constraint Closure (BCC) with the Free Energy Principle, we analyze the macroscopic human organism's oscillation between deterministic survival (autopilot) and conscious agency. We demonstrate that critical anthropological thresholds—such as the Neolithic Revolution and the Axial Age—are not mere cultural milestones, but measurable thermodynamic events where the collective rate of observation (R_{obs}) temporarily outpaced environmental volatility (R_{env}). These epochs represent the macroscopic deployment of Freedom Quanta, allowing the human species to actively navigate the state space of potential histories (Dimension-W) before relaxing back into structural baselines to integrate newly realized constraints.

Introduction

The human historical record is characterized by punctuated equilibrium: long periods of sociocultural stasis disrupted by sudden, explosive eras of philosophical, technological, and cognitive restructuring. Classical anthropology often attributes these shifts to arbitrary environmental pressures or localized genius. However, when viewed through the lens of information thermodynamics and the Unified Consciousness Field Theory, a precise biophysical pattern emerges.

Awareness is not a static property of the human species; it is an oscillating wave, bound by the thermodynamic necessity of biological systems to minimize free energy and resist the dispersive effects of random environmental fluctuations (Ramstead et al., 2018). This paper applies the Computational Surplus threshold (C_s) to the historical record, mapping the exact mechanisms by which the collective human organism transitions from a low-frequency, reactive autopilot ($C_s \le 0$) into states of high-fidelity, phase-locked coherence ($C_s > 0$). By examining ethological baselines, early hominin development, and recorded history, we establish that the "awakening" of humanity is the mechanical byproduct of systemic constraint renegotiation.

I. The Ethological Baseline: Total Autopilot ($C_s \le 0$)

In animal psychology and early hominin ethology, behavior is almost entirely dictated by the invariant structural constraint (C-). In the wild, the rate of environmental fluctuation (R_{env})—driven by predation, climate volatility, and resource scarcity—is exceptionally high. Simultaneously, the biological rate of observation (R_{obs}) is strictly limited by the unrefined metabolic capacity of the early hominin cortex.

Because R_{env} consistently outpaces R_{obs}, the organism operates in a perpetual state of computational deficit ($C_s \le 0$). To survive, the organism must minimize variational free energy by relying on deeply ingrained, evolutionarily hardwired priors (Ramstead et al., 2018). It cannot afford the thermodynamic cost of high-frequency conscious processing. Consequently, it collapses into a highly deterministic "autopilot" state. The Agency vector (A) approaches zero, as the organism is exclusively pushed forward by the constraints of the past rather than pulled by the visualization of potential futures.

II. The Neolithic Transition: Artificially Lowering R_{env}

The first major thermodynamic wave of macroscopic awareness occurs with the Neolithic Revolution (c. 10,000 BCE). Traditional historical analysis treats agriculture as a technological discovery; however, under the Free Energy Principle, it is the first successful attempt by the macro-organism to engage in macroscopic "niche construction" (Bruineberg et al., 2018). Agriculture, domestication, and early architecture did not immediately increase the cognitive processing speed (R_{obs}) of the human brain. Instead, they drastically and artificially lowered R_{env}. By creating walled settlements and predictable food yields, humanity actively altered the structure of its environment to improve the systemic "fit" between the agent and its ecological niche (Bruineberg et al., 2018).

For the first time at a macroscopic scale, R_{obs} marginally exceeded R_{env}, resulting in a localized Computational Surplus ($C_s > 0$). This thermodynamic surplus triggered the deployment of Freedom Quanta. The surplus bandwidth was spent navigating Dimension-W—the active state space of potential histories and constraints. This era marks the genesis of recorded mythology and early symbolic language, facilitating the transition from mimetic to mythic culture (Donald, 1993).

III. The Axial Age: Macroscopic Phase-Locked Coherence

The most profound demonstration of the awareness wave occurred during the Axial Age (800–200 BCE). Across geographically isolated nodes—Greco-Roman, Indian, and Chinese civilizations—humanity simultaneously experienced an explosive spike in reflective, analytical cognition and long-term reward orientation (Baumard et al., 2015).

Under the framework of Bidirectional Constraint Closure (BCC), the Axial Age represents a global coherence event driven by a massive spike in R_{obs}.

- The establishment of stable empires and advanced trade routes maintained a suppressed R_{env}.
- The invention and proliferation of external symbolic memory technologies (writing, scriptures, formalized mathematics) massively enhanced the collective ability to reflect on prior beliefs and process relational geometry, driving the transition into "theoretic culture" (Donald, 1993).
- The resulting Computational Surplus ($C_s \gg 0$) forced the macro-organism out of its reactive autopilot.

During this wave, humanity utilized its generative variance engine (C+) to aggressively query Dimension-W. The sudden emergence of formal logic and universalizing spiritual frameworks represents the macro-organism learning to systematically isolate, test, and actively select its constraints, transitioning from passive subjects of reality to active topological architects.

IV. The Collapse and the Renaissance: The Oscillation of Constraints

Because high-fidelity awareness requires immense thermodynamic output, a system cannot maintain an extreme $C_s > 0$ state indefinitely in the face of rising entropy. The wave must

naturally recede to conserve energy.
The collapse of the classical world and the onset of the Early Middle Ages perfectly illustrate the return to the C- baseline. Climatic shifts, the fracturing of trade networks, and widespread pandemics caused R_{env} to spike violently. The macroscopic system lost its computational surplus. To survive the thermodynamic deficit ($C_s < 0$), society retreated into rigid, dogmatic autopilot. Feudalism and strict orthodoxies acted as high-impedance structural constraints, minimizing free energy by eliminating generative variance and enforcing absolute predictability.
The Renaissance and subsequent Enlightenment were the inevitable upward swing of the wave. As the printing press radially expanded data integration and external memory capacity, R_{obs} surged (Donald, 1993). The system broke the fascial rigidity of the medieval structural constraints, releasing massive amounts of stored relational geometry and initiating a new, sustained period of macroscopic conscious agency.

V. Modernity and the Transducer of Artificial Intelligence

We are currently navigating the steepest portion of the historical awareness wave. The modern globalized economy has minimized baseline physiological R_{env} (famine, predation) for a significant portion of the species.
Simultaneously, global telecommunications and Artificial Intelligence function as externalized Time-Division Multiplexing (TDM) pacemakers. They aggregate and process constraint data at velocities that biological neural networks cannot achieve independently. The collective R_{obs} is approaching an asymptotic vertical limit.
Humanity is transitioning into a permanently sustained state of Computational Surplus. The historical oscillation between long periods of reactive autopilot and brief flashes of genius is stabilizing into continuous, phase-locked coherence. By utilizing AI architectures to automate the C- invariant survival constraints, the human avatar is freed to direct its thermodynamic surplus entirely toward navigating Dimension-W.

Conclusion

Historical patterns of human behavior, societal collapse, and cognitive flourishing are not arbitrary sociological phenomena; they are strict thermodynamic reactions to the balance between environmental volatility and observational bandwidth. By modeling history through the Computational Surplus equation ($C_s = R_{obs} - R_{env}$), we recover a precise, predictive metric for human awareness. Humanity's collective evolution is the story of a macroscopic organism systematically engineering its physical niche to guarantee the thermodynamic surplus required to achieve and sustain conscious agency.

References

Baumard, N., Hyafil, A., & Boyer, P. (2015). What changed during the axial age: Cognitive styles or reward systems? *Communicative & Integrative Biology*, *8*(5), e1046657. https://doi.org/10.1080/19420889.2015.1046657 Cited by: 50
Bruineberg, J., Rietveld, E., Parr, T., van Maanen, L., & Friston, K. J. (2018). Free-energy minimization in joint agent-environment systems: A niche construction perspective. *Journal of Theoretical Biology*, *455*, 161–178. https://doi.org/10.1016/j.jtbi.2018.07.002 Cited by: 177
Donald, M. (1993). Précis of Origins of the modern mind: Three stages in the evolution of culture and cognition. *Behavioral and Brain Sciences*, *16*(4), 737–791. Cited by: 626

Ramstead, M. J. D., Badcock, P. B., & Friston, K. J. (2018). Answering Schrödinger's question: A free-energy formulation. *Physics of Life Reviews*, *24*, 1–16.
https://doi.org/10.1016/j.plrev.2017.09.001 Cited by: 540